

Day 1 lab: toolchain, sensor tests, and model budgets

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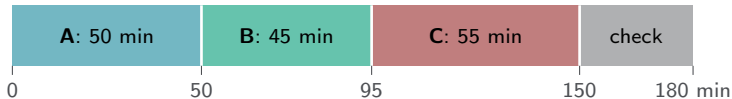
- ▶ **Goal:** your group has a toolchain that works, and a first resource budget for each device class.
- ▶ **Board:** the XIAOMI Kit. Each group has two students and one kit.
- ▶ **Deliverable:** the file `report.md` with the results of the five sensor tests, the completed table “model and device”, and the Decision Log.

From the lecture to the lab

The lecture gave the method: three budgets of a model, and two budgets of a device. In the lab, you read the budgets of your board, and you calculate the budgets of three models.

Key point: You decide from flash and RAM numbers if a model fits a device.

Plan of the three hours



Part	Work	Time
A	Toolchain: Arduino IDE, esp32 core, Python environment, Blink	50 min
B	Sensor tests: IMU, display, microphone, camera, Wi-Fi	45 min
C	Model budgets: memory of the board, notebook, table, Decision Log	55 min
Check by the instructor		30 min

Write each result in `report.md` when you get it. Do not wait for the end of the lab.

Hardware for each group

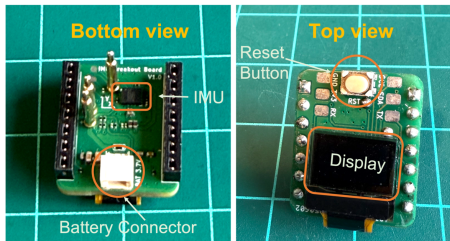
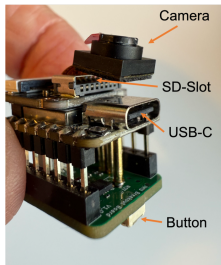
- ▶ XIAOML Kit: the XIAO ESP32S3 Sense with the camera, and the expansion board with the IMU and the display.
- ▶ USB-C data cable. A cable for charging only does not work.
- ▶ Wi-Fi antenna of the kit.
- ▶ Laptop: Windows, macOS, or Linux.

Lab files	Labs/day01/
Setup steps	Labs/SETUP.md
Tool versions	Labs/VERSIONS.md

Software

- ▶ Arduino IDE 2.
- ▶ Board core: esp32 by Espressif Systems 3.3.12.
- ▶ Library Seeed Arduino LSM6DS3 2.0.7 (IMU).
- ▶ Library U8g2 2.36.19 (display).
- ▶ Python 3.10 or later with torch, torchvision, numpy, matplotlib, and jupyterlab.

The kit has two boards and no wires



- ▶ The XIAO sits on the expansion board. You connect no wire.
- ▶ The expansion board has the IMU (I2C address 0x6A), the display (I2C address 0x3C), and the reset button.
- ▶ Connect the USB-C cable to the XIAO. The cable gives power and the serial port.
- ▶ Do not install the heat sink. It does not fit under the expansion board.

Source: Photographs and data: XIAO ML Kit setup chapter of *Machine Learning Systems*.

1. Do steps 1 to 4 of Labs/SETUP.md: the Arduino IDE, the esp32 core, the two libraries, the board settings, and the Python environment. On Linux, do also step 9 (permission for the USB port).
2. Connect the kit. Click Select Board, enter xiao, and select XIAO_ESP32S3 and the port of the board.
3. Set Tools > PSRAM > Disabled.
4. Open sketches/blink/blink.ino. Click Upload.
5. Start Jupyter. Run only section 0 (Setup) of the notebook.

```
python3 -m venv .venv  
.venv/bin/pip install -r Labs/requirements.txt  
.venv/bin/jupyter lab Labs/day01/model_budgets.ipynb
```

You see: the versions of Python, numpy, torch, and torchvision, and no error.

On Linux, install the CPU version of PyTorch first. The README gives the command. On Windows, the programs are in .venv\Scripts\.

You see: the built-in LED is on for one second and off for one second. The Serial Monitor at 115200 baud prints LED on and LED off.

Read the last lines of the build output in the Arduino IDE:

```
Sketch uses 271701 bytes (8%) of program storage space.  
Maximum is 3342336 bytes.  
Global variables use 21824 bytes (6%) of dynamic memory,  
leaving 305856 bytes for local variables. Maximum is 327680 bytes.
```

Flash	 sketch: 271 701 bytes (8 %)	3 342 336 bytes
-------	---	-----------------

RAM	 global variables: 21 824 bytes (6 %)	327 680 bytes
-----	--	---------------

- ▶ The first line is the flash budget of a sketch. The second line is the RAM budget with no PSRAM.
- ▶ **You record:** your two lines. Your numbers can be different.

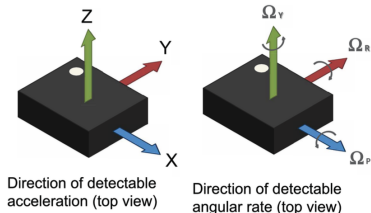
Source: Result of the compiler for blink with the esp32 core 3.3.12. The IDE prints two lines. The frame shows each line on two lines.

Part B: five sensor tests (45 min)

Test	Sketch	PSRAM	You see
IMU	<code>imu_test</code>	Disabled	Six values, ten times each second, in the Serial Monitor
Display	<code>oled_test</code>	Disabled	Hello and World! in a frame on the display
Microphone	<code>mic_test</code>	Disabled	A line in the Serial Plotter that follows your voice
Camera	<code>camera_test</code>	OPI PSRAM	One line for each frame, each second
Wi-Fi	<code>WiFiScan</code> (example)	Disabled	The list of the networks with their signal strength

- ▶ The first four sketches are in `Labs/day01/sketches/`.
- ▶ Use about 8 minutes for each test. The serial speed is 115200 baud.

Key point: These five sensors give the data of the later labs. A sensor that fails today stops a later lab.



Test 2: display

- ▶ Upload `oled_test`.
- ▶ **You see:** the text Hello and World! in a frame on the display of the expansion board.

Test 1: IMU

- ▶ Upload `imu_test`. Open the Serial Monitor.
- ▶ **You see:** IMU initialized successfully, then six values: the acceleration on three axes and the angular rate around three axes.
- ▶ Put the kit flat on the table: Z is near 1.0 g. X and Y are near 0.
- ▶ Turn the kit on its side: a different axis shows 1.0 g or -1.0 g.
- ▶ **You record:** the Z value with the kit flat.

Source: Figure and expected values: XIAOML Kit setup chapter of *Machine Learning Systems* (CC BY-NC-SA 4.0).

Test 3: microphone

- ▶ Upload `mic_test`. Close the Serial Monitor. Open Tools > Serial Plotter.
- ▶ **You see:** small values in silence. Speak or clap: the line shows large positive and negative values.

Test 4: camera

Predict before the test. One frame has 320×240 pixels with one byte for each pixel. How many bytes has one frame? Does one frame fit in the RAM of a microcontroller with 256 KB?

- ▶ Set Tools > PSRAM > OPI PSRAM. Upload `camera_test`.
- ▶ **You see:** Camera initialized, then one line each second with these five values: `width,height,bytes,capture_ms,mean_brightness`
- ▶ Cover the lens: the last value decreases. A lamp: it increases.
- ▶ **You record:** the bytes of one frame and the capture time.



Test 5: Wi-Fi

- ▶ Connect the antenna to the small connector on the XIAO. Put one side of the antenna connector in first. Then press the other side down.
- ▶ Open File > Examples > WiFi > WiFiScan. Upload it. The example needs no password.
- ▶ **You see:** a list of this form.

```
Scan start
Scan done
2 networks found
Nr | SSID                | RSSI | CH | Encryption
1 | ...                  | ... | .. | ...
```

- ▶ **You record:** the number of networks and the RSSI of the strongest network.

RSSI is the signal strength in dBm. A value nearer to 0 is a stronger signal.

Source: Photograph: XIAO ML Kit setup chapter of *Machine Learning Systems* (CC BY-NC-SA 4.0).

Part C: the memory of the board (10 min)

1. Set Tools > PSRAM > Disabled. Upload sketches/memory_report. Record the three values of the heap.
2. Set Tools > PSRAM > OPI PSRAM. Upload again. Record the same three values and the two values of the PSRAM.

```
--- Internal RAM (heap) ---
Heap size                ... bytes    ... KB    <- record
Free heap                ... bytes    ... KB    <- record
Lowest free heap since start ... bytes    ... KB
Largest block that malloc can give ... bytes    ... KB    <- record
--- External RAM (PSRAM) ---
PSRAM size               ... bytes    ... KB    <- record
Free PSRAM               ... bytes    ... KB    <- record
```

- ▶ “Largest block that malloc can give” is your RAM budget with no PSRAM. A model needs one block for all its tensors.
- ▶ “Free PSRAM” is your RAM budget with PSRAM.

Key point: You measure the budget. The 8 MB of the data sheet is not the free memory.

Source: Form of the output of `memory_report`. The marks “<- record” are not in the output. Nobody measured the values on a board.

Open `model_budgets.ipynb`. Each task has the mark TODO (student).

Task	Work	Notebook
1	Write the function <code>conv_cost</code> : the cost of one convolution layer	Section 1
2	Write your predictions for three models	Section 3
3	Put your two measured RAM budgets of the XIAO in the device table	Section 5
4	Write the function <code>fits</code> : does a model fit a device?	Section 5
5	Write the Decision Log	Section 6

The check cell of task 1 compares your function with PyTorch:

```
standard, stride 2: correct
  expected (48, 48, 448, 995328)
  yours    (48, 48, 448, 995328)
...
Task 1: complete
```

The four numbers are: output height, output width, parameters, MACs. The complete notebook runs in less than one minute on a laptop CPU.

`profile_model` gives the three budgets of a model. For the example network of the lecture: 24 234 parameters, 6 304 384 MACs, and a peak of 64 512 values.

Model	Input	Design
MobileNetV2	$224 \times 224 \times 3$	Inverted residual blocks, for phones
ResNet-18	$224 \times 224 \times 3$	Standard convolutions with skip connections
Small depthwise CNN	$96 \times 96 \times 3$	Depthwise separable blocks, for a microcontroller

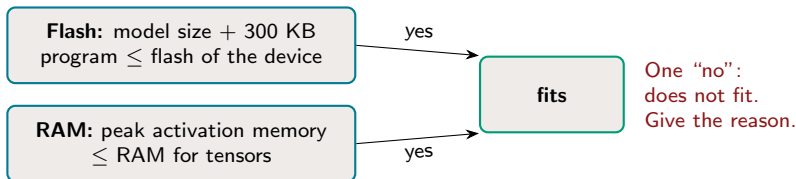
Task 2: predict before you run section 4.

- ▶ Which model has the most parameters?
- ▶ Which model has the most operations (MACs)?
- ▶ Which model has the largest peak activation memory?
- ▶ What is the ratio of the parameters, ResNet-18 to MobileNetV2?

A wrong prediction costs no points. A missing prediction does.

Source: Models: torchvision (BSD-3-Clause).

Part C: does the model fit the device?



Device	Flash	RAM	Source of the RAM value
Microcontroller, 256 KB	1.0 MB	211.1 KB	256 KB less the static RAM of a sensor sketch
XIAO, no PSRAM	3.2 MB	298.7 KB	Result of the compiler. Task 3: your "largest block".
XIAO, with PSRAM	3.2 MB	8.0 MB	Size of the chip. Task 3: your "free PSRAM".
Raspberry Pi 5	64 GB	8 GB	microSD card and RAM of the board

You record: six rows (three models in `float32` and in `int8`) for four devices. Each "does not fit" has a reason: flash, RAM, or flash and RAM.

Source: Device budgets of the notebook `model_budgets.ipynb`. The column "RAM" is the RAM for the tensors. 1 KB = 1024 bytes.

The instructor checks this at the end of the lab:

- ▶ ✓ All five sensor tests of Part B pass.
- ▶ ✓ The report has the two lines of the build output, and the values of `memory_report` for the two PSRAM settings.
- ▶ ✓ The notebook prints Task 1: complete.
- ▶ ✓ The report has the predictions of task 2. You wrote the predictions before the measurement.
- ▶ ✓ The table has six rows and four devices. Each “does not fit” has a reason.
- ▶ ✓ The Decision Log gives numbers and names one trade-off.

Hand in: the file `report.md`. Show the notebook and the report to the instructor.

Key point: A result with no reason is not complete. “Does not fit” is the start of the answer. The budget that decides is the answer.

Write about 100 words. State one design decision, give your measured numbers, and name the trade-off.

You must classify images of 96×96 pixels on the XIAOML Kit.

- ▶ Which of the three models do you select?
- ▶ In which data type: `float32` or `int8`?
- ▶ With PSRAM or with no PSRAM?
- ▶ Which budget decides?

A good Decision Log:

- ▶ gives the model size and the peak activation memory from your table,
- ▶ gives the flash budget and the RAM budget that you measured on your board,
- ▶ names what you lose with your decision. Example: memory against accuracy.

Problem	Fix
The Arduino IDE shows no port	Use a data cable. On Linux, do step 9 of Labs/SETUP.md.
The upload fails with a connection error	Disconnect the cable. Hold the B (boot) button of the XIAO. Connect the cable, then release the button. Upload again.
The Serial Monitor shows nothing or wrong characters	Set 115200 baud.
<code>imu_test</code> prints IMU initialization failed	Press the XIAO on the expansion board until all pins are in.
<code>mic_test</code> prints nothing	Open the Serial Plotter or the Serial Monitor. The sketch waits for it.
<code>camera_test</code> prints PSRAM is not active	Set Tools > PSRAM > OPI PSRAM and upload again.
WiFiScan finds no network	Connect the antenna to the small connector on the XIAO.
The notebook prints No module named 'torch'	Start Jupyter with <code>.venv/bin/jupyter lab</code> .

The README of the lab has the complete list. Ask the instructor when a part takes more than the planned time.

This lecture adapts material from these sources:

- ▶ *Machine Learning Systems* by Vijay Janapa Reddi and contributors, Harvard University. mlsysbook.ai, CC BY-NC-SA 4.0.
- ▶ *TinyML Made Easy: XIAO ESP32S3* by Marcelo Rovai. github.com/Mjrovai/XIAO-ESP32S3-Sense, Apache-2.0.

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